

Recall, not verification, is the bottleneck when frontier LLMs elucidate molecular structures from real spectra

Ilkham Yabbarov^{a*}, Rudra Sondhi^a, Rodrigo A. Vargas-Hernández^a

^a Department of Chemistry and Chemical Biology, McMaster University, Hamilton, Ontario L8S 4L8, Canada.

*E-mail: yabbaroi@mcmaster.ca

Given the molecular formula, the infrared band list and the $^1\text{H}/^{13}\text{C}$ shift lists exactly as reported in an open-access paper, a frontier large language model recovers the correct molecular constitution for 28% of 194 compounds (top-1; 95% CI 22–35) — far below the near-100% implied by curated demonstrations, and 15% once the benchmark’s deliberate 50/50 difficulty balance is reweighted to the corpus it was drawn from. The bottleneck is proposal, not judgment: the true structure is generated for only 34% of compounds, and where it is, a training-free forward-verification step that predicts each candidate’s ^{13}C spectrum and re-ranks by agreement with experiment selects it 89% of the time (58/65; 81% on the 37 compounds where a ranker had a choice). The same split, recovered from published top- k figures, shows that recall carries 68–83% of the accuracy collapse when three independent systems move from curated or simulated data to real, heterogeneous spectra. Generating more candidates lifts recall from 32% to 42% and top-1 from 23% to 30% on 60 compounds; verification itself moves whole-benchmark top-1 only from 28% to 30%. Masking the spectra drops top-1 from 23% to 5%, and Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol all verify better than they generate. We release *IRexp*, the largest openly redistributable collection of *experimental* infrared band lists (121,233 records), *IRSpectra-Bench*, a blind mechanically scored evaluation of 194 compounds, and all predictions and code.

1. Introduction

Determining structure from spectra is central to synthetic and analytical chemistry. Machine learning uses encoder–decoder models on paired corpora; *Spectro*, for one, learns $^1\text{H}/^{13}\text{C}/\text{IR} \rightarrow \text{SELFIES}$ from 6,833 molecules¹. General-purpose LLMs are now reported off-the-shelf: a non-peer-reviewed 2026 industrial white paper found Claude Opus matching or beating commercial NMR-prediction software forward (structure $\rightarrow$ spectrum, ± 0.08 ppm ^1H) and “recovering all eight simpler structures on every attempt” in the inverse². We treat those numbers as a claim to test, not an established baseline.

That evaluation is narrow: 15 inverse problems on curated single-ring or two-fragment molecules, NMR only, seven given the *starting-material structure* as a hint, and “recovery” scored leniently over three runs and three ranked candidates. The chemist’s question — *take an arbitrary experimental spectrum from a paper and recover the structure* — remains open. It needs a large, diverse, real benchmark; blind, reproducible scoring; and honest accounting for choices that inflate or deflate the score.

We provide all three. *IRexp* (§2) is, to our knowledge, the largest openly *redistributable* collection of experimental IR *band lists* (121,233 records; 43,060 structure-linked; 33,201 IR + ^1H + ^{13}C + structure); view-only libraries such as SDBS hold more structure-linked *spectra* but cannot be redistributed (§2.1). *IRSpectra-Bench* (§3) is a blind, mechanically scored, complexity-stratified

evaluation on real spectra — not the first such benchmark (MolQuest scores 530 post-2025 compounds by exact canonical SMILES³) — but ours adds stratification, multimodal IR + ¹H + ¹³C input, and decomposition: measured on Claude, replicated across three other model families (§4.7), with a within-compound solver-methodology control (§4). Ground truth is literature structures resolved deterministically (OPSIN/RDKit) and checked mechanically (560/560 bands on a seed-fixed sample, §2.3; expert-chemist review prepared but not yet run, §8); no LLM curates labels or scores predictions.

Forward-verification elucidation (§5) turns the model’s *strong* direction (forward prediction) against its *weak* one (inverse regiochemistry); the loop is prior art (§1.1), the decomposition ours, run training-free on every compound. The finding is sharp asymmetry: given a candidate set containing it, the model *verifies* the correct structure 89% of the time yet *proposes* it for only 34%, so generation binds the result (Fig. 1). Gain is bounded (top-1 28% → 30% on n=194; 23% → 30% on the 60-compound arm where wide generation was also tested) and, by our own measurements, cannot exceed a recall/precision ceiling without sharper verification or 2D-NMR data.

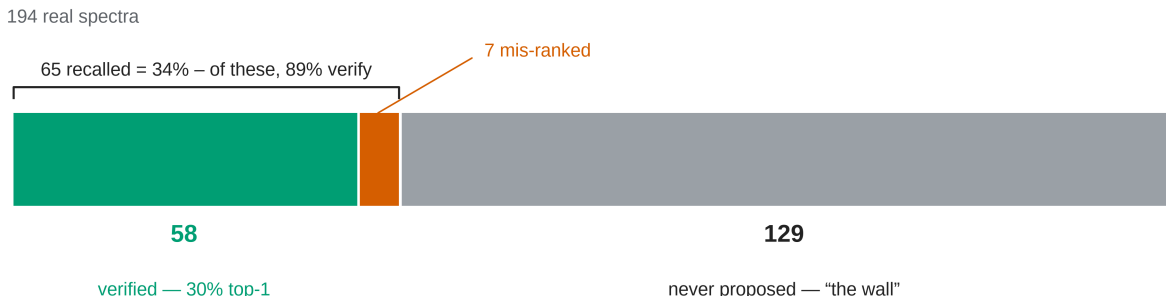


Fig. 1. All 194 compounds as one part-to-whole bar: verified top-1 for 58 (green), recalled but mis-ranked for 7 (vermilion), never proposed for 129 (grey) — *the wall*, 66% of the bar. Generation recall is 65/194 = 34%; of those 65, forward-verification ranks 58 first (58/65 = 89% precision conditional on recall). Their product is the 30% end-to-end top-1. The two rates have different denominators and are not differenced.

Solver and verifier (Fig. S1) are LLM agents on a consumer subscription, with no fine-tuning and no API spend — cheap to *re-run* but not reproducible: no pinned snapshot, temperature or seed, and an undisclosed product system prompt. Our dispatch instruction for the main rounds was not captured, only the per-compound data it wrapped, so outsiders reproduce this distributionally (§9). Scoring is exactly reproducible: predictions, ground truth and scorers are released, and every number in the training-free core (§3–§5.3, §4.6) regenerates from them. Exceptions are §5.4’s HOSE lookup and GNN, which need an nmrshiftdb2 dump we cannot redistribute (*Data availability*); that GNN and the §5.6 generator are trained probes, fenced as complements to the training-free core.

1.1 Related work

Trained spectra → structure models. The dominant line trains sequence or graph decoders on spectra: *Spectro*, from our own group¹; a multitask CNN+transformer for routine 1D-NMR⁴; set/graph transformers such as NMRTrans⁵; and — closest to our multimodal setting — *NMIRacle*⁶, conditioned jointly on IR + ¹H + ¹³C. They are accurate *in-distribution* but retrained per modality and need a labelled spectra → structure corpus. One concurrent effort attacks that scarcity as we do: NMRTrans builds NMRSpec, a literature-mined corpus of experimental ¹H/¹³C spectra⁵. IReXP is the complement — infrared band lists under permissive redistribution licences (§2.3) — and the two together are the open, experimental, multimodal corpus neither supplies alone. Our group’s prior

infrared work sits upstream: *vIR-OLO* recasts functional-group identification as object detection, localising bands as well as labelling them⁷, and *j-IR-vis* learns transferable spectral embeddings from IR⁸. Neither generates candidate structures, the step this paper finds binding.

LLMs as elucidators, and how this work differs. LLM-orchestrated tools already plan and execute real syntheses (ChemCrow⁹, Coscientist¹⁰); such systems *call* an elucidation routine, and this paper measures what one returns. General-purpose LLMs have been applied off-the-shelf², as have multimodal models over multi-spectral input (SpectraLLM, SpecMol)^{11,12} and knowledge-enhanced tree-search reasoning¹³; *MolPuzzle*¹⁴ supplies IR+MS+¹H+¹³C puzzles with the formula given, and *IR-Agent*¹⁵ emulates expert IR interpretation on experimental spectra with a multi-agent framework. Priessner *et al.* run the same generate-then-rerank shape in this journal, re-ranking generator candidates with a reasoning LLM¹⁶. We claim priority on none of them: IR-Agent improves *how the model reads a spectrum*, where we ask what limits the outcome once it has.

Prior benchmarks and trained baselines evaluate on simulated or software-predicted spectra^{1,6} or on curated single-instrument libraries — Alberts *et al.* use NIST gas-phase IR^{17,18} — where IRSpectra-Bench uses spectra as reported in source papers, across thousands of laboratories and instruments. Recent benchmarks bound the task elsewhere: NMRGym’s 269,999-molecule scaffold-split experimental ¹H/¹³C suite¹⁹; MolQuest, agentic and multi-turn, where the best models reach roughly 50% and most fall below 30%³; and SpecX, where multimodal LLMs “lack precise spectral grounding”²⁰ — the limit §5.1 measures. That line is also the state of the art for *trained, formula-conditioned IR* elucidation — an IR → structure transformer¹⁷ now at 63.8% top-1 / 84.0% top-10 on experimental NIST gas-phase spectra given the formula, pretrained on 1,399,806 simulated spectra¹⁸. That headline is on a 6–13-heavy-atom subset, the range where our own accuracy is highest (60.5% top-1 for ≤15 heavy atoms, §4.1), so on comparable molecules the training-free LLM and the purpose-trained transformer are closer than the headline numbers suggest. The same work reports 59.94% on a 5–35-heavy-atom set (n=5,024), spanning most of our range and falling only four points below that 63.8% headline — a single-instrument gas-phase library against heterogeneous literature-reported band lists. Neither setting bounds the other. Reported accuracies also swing with inference method and scoring harness: GPT-4o is scored at 1.4% on MolPuzzle by the benchmark’s own authors¹⁴, at 27.8% under a plain chain-of-thought harness, and at 57.8% with knowledge-enhanced tree-search reasoning¹³ — a factor of forty for one model on one benchmark. Cross-paper numbers are therefore not comparable, and we fix and release a single pre-registered RDKit-InChIKey protocol with bootstrap CIs.

Computational NMR for structure validation. Forward-verification is the LLM analog of a workflow chemists already trust — compute the spectrum each candidate *would* give and match it to experiment: DP4 / DP4+ over GIAO-DFT shifts in solution^{21,22}, *NMR crystallography* with GIPAW shifts in the solid state^{23,24}. We swap the quantum-chemical predictor for a forward LLM, trading accuracy for zero setup cost, and inherit the principle that *verification by forward prediction is easier than inverse generation*. *Computer-assisted structure elucidation* (CASE) has run the loop since the 1990s and overturned published assignments^{25,26}; its generator is a strict enumerator with exhaustive recall by construction²⁶ — the axis on which we find the LLM weak, and the axis the LLM benchmark literature almost never reports: across every paper we could read in full it appears once, in a supplementary figure caption (§6). The decomposition is stable under every perturbation we could run — four Claude models, a second chemical domain, four verifiers, all recall-bound (§4.4, §4.5, §5.4). A four-vendor replication on 60 compounds finds the same inequality outside the Claude family (§4.7); the headline n=194 result remains a single vendor (§8).

The generate-and-forward-verify loop is not ours. *NMR-Solver*²⁷ builds it without an LLM, retrieving and fragment-recombining candidates and ranking them by $^1\text{H}/^{13}\text{C}$ shifts forward-predicted with NMRNet (^{13}C MAE 1.098 ppm); on ≈ 450 experimental literature spectra with the formula supplied it reports 52.89% top-1 with stereochemistry ignored, well above our 28.4% on the same criterion; the twin supplementary table scores the identical predictions with stereochemistry preserved and gives 31.56%, against our 21.1%. We quote both because the gap between them, 21 points on one system’s own predictions, is larger than most differences anyone compares. §5 asks a different question — *how much of the remaining error is generation and how much is verification* — answered by decomposition (§5.2) rather than by a better solver. Its predictor is roughly twice as sharp as the ≈ 2 ppm LLM forward-predictor that §5.1 identifies as the binding constraint, and it delivers roughly twice the top-1. That is what our §5.4 ablation predicts and what §5.1’s separability measurement implies. *NMRAgent*²⁸ couples spectral tools to a knowledge graph and validates on newly isolated natural products; it is complementary — a best-effort system, where we measure an off-the-shelf model.

Concurrent work, and what we do not claim. Kliuev *et al.* give six frontier LLMs the ^1H and ^{13}C peak lists for 105 molecules: the best solves 24%, against 69% for the best of four specialised solvers, and every model “drift[s] toward smaller molecules”²⁹ — an independent replication of our headline and of its size dependence. Espejo Morales *et al.* instead put a frozen LLM in a curated environment of processing tools, tabulated shifts and hard formula, unsaturation and integration checks: 80.9% top-1 on a chemistry-education set ($N=236$), 71.1% on the 15 molecules where graduate students reach 66.7%, and 20.6% on 34 industrial samples where a trained transformer reaches 14.7%³⁰. Their frame — search rather than a learned map — is orthogonal to ours: they ask which architecture to adopt, we ask which stage of it binds. Their inputs are raw instrument files and ours are published reports, which they exclude by design (§8). We claim priority on neither that frame, nor on external simulation tools improving agent elucidation, nor on placing frontier LLMs near a quarter on real peak lists. Nor on the idea that candidate supply binds: Priessner *et al.* measure generator recall as a standalone quantity and state the constraint outright — a re-ranker “cannot overcome cases where the correct structure is entirely absent from the candidate pool”¹⁶. What we add is the measurement at scale and the generalisation: the split on 194 compounds across four model families and four verifiers, recovered from the published top- k figures of every system we could read (§6), where it turns out to hold in some regimes and fail in others.

2. The IRexp dataset

2.1 Motivation

An IRexp record is a *band list* — peak positions in cm^{-1} transcribed by an author into a paper’s text — with co-reported $^1\text{H}/^{13}\text{C}$ shift lists and, where resolvable, the 2D structure. IRexp contains no absorbance traces. That is the published form a language model consumes, but a different object from a digitised spectrum, and the two should not be counted against one another.

Among *digitised spectra*, free downloads run to the NIST WebBook³¹ ($\approx 16\text{k}$) and the Chemotion ELN deposit³² ($\approx 2\text{k}$). AIST SDBS³³ is larger ($\approx 54\text{k}$ FT-IR, all structure-linked) but *view-only* (50 spectra/day, no bulk export), so IRexp does not compete there: SDBS alone holds more structure-linked spectra than IRexp holds structure-linked band lists. Among text-derived *band lists* IRexp is the largest *openly redistributable* collection by record count — deliberately scoped to that object type.

2.2 Construction

Experimental sections report per-compound IR band lists with co-reported $^1\text{H}/^{13}\text{C}$ NMR in a stable textual convention. Mining it is mature³⁴; we add a pipeline specialised to the *IR band list*, the modality those tools passed over.

- **Discovery.** Open-access literature is harvested in bulk from the PMC Open-Access Subset³⁵ and the Chemotion FT-IR deposit (RADAR4Chem).
- **Extraction.** A deterministic parser segments experimental text per compound and extracts IR wavenumbers and $^1\text{H}/^{13}\text{C}$ shifts, gated against scan-range artefacts and prose false-positives.
- **Resolution.** IUPAC names go to SMILES with OPSIN³⁶ and RDKit³⁷ canonicalisation (InChIKey, SELFIES³⁸), with a PubChem³⁹ fallback for trivial names; handling open-access label conventions and narrative artefacts raised structure coverage from 24% to 35%.

2.3 Contents and licensing

Table 1 summarises contents and provenance.

Table 1. IRExp dataset contents and provenance.

field	value
experimental IR band-list records	121,233
...co-reporting ^1H and/or ^{13}C NMR	87,075 (72%)
...with a resolved 2D structure	43,060 (35.5%)
...of these, with ^1H and/or ^{13}C NMR	40,702
...of these, with both ^1H and ^{13}C (full quadruples)	33,201
<i>by provenance:</i> author-transcribed from PMC-OA text (CC-BY)	119,345
<i>by provenance:</i> peak-picked from Chemotion ELN deposits (CC-BY-SA)	1,888

The pools differ: PMC records are author-transcribed (median 9 bands), Chemotion records peak-picked from deposited spectra (median 39); users training on band density should treat them apart. `scripts/split_license_pools.py` separates them on `source_doi` and stamps each record’s licence.

Extraction fidelity is measured. On a seed-fixed random sample of 60 PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`) we re-fetched each article and checked every recorded wavenumber against its text: 560/560 bands and 60/60 records were confirmed (Wilson 95% CI 99.3–100% and 94–100%). This bounds *transcription* error — hallucinated, mis-parsed or unit-mangled values — below 1% of bands. It does not measure whether the parser found every IR string in every paper — a recall question requiring human reading; that audit is prepared but not yet run (§8).

`irexp_resolved` (43,060 records, 100% structure-linked) is the benchmark-ready split, $\approx 6\times$ the 6,833-molecule set used to train Spectro¹ (Fig. S2).

3. Benchmark design (IRSpectra-Bench)

From `irexp_resolved` we draw *IRSpectra-Bench*, 194 blind elucidation problems, each giving the *molecular formula* (as from HRMS), the *IR band list* and the ^1H and ^{13}C *shift lists*; no name, SMILES or scaffold hint. In 10 of the 194 the authors’ peak assignments name a ring system (2CH₂·pyrrolidine, pyridazinone H5). Those compounds are solved 1/10 against 54/184 (29.3%) for the rest (`scripts/prompt_leakage.py`), so the annotation does not carry the headline; we keep them rather than drop them. Main-round ground truths are spectrally validated by an automated RDKit check (^{13}C peaks vs symmetry-unique carbons, formula match, SELFIES round-trip) excluding merged or incomplete spectra — 6/140, leaving 134; `scripts/validate_benchmark.py` applies the same filter to all three rounds, regenerating every `clean_qids.json` from the released questions and answers.

The filter does not gate on ^1H . In 13 of the 194 retained records the reported ^1H integral exceeds the reference structure’s hydrogen count — residual solvent, water, exchangeable protons, or a rotamer mixture — printed as a diagnostic, not excluded; the cohort was fixed in advance, and re-filtering post hoc on a criterion chosen after seeing results is a degree of freedom a benchmark should not take. We report the sensitivity instead: dropping all 13 moves the headline from 28.4% to 29.3% (53/181), +0.9 points, leaving every conclusion unchanged. The controlled rounds’ 60 compounds are fixed, pre-registered sets, audited the same way but reported rather than filtered (57/60 pass; §8).

Difficulty is declared in advance, as a property of the compound. By RDKit ring analysis a compound is *simple* iff it has at most two rings, no fused/spiro/bridgehead system and ≤ 22 heavy atoms; all others are *complex* (98 simple / 96 complex). The threshold is not load-bearing: sweeping it from 18 to 26 leaves the simple-minus-complex top-1 gap inside a 36–40-point band throughout (36.1 at the narrowest, 39.6 as released; `scripts/difficulty_sensitivity.py`). This axis is distinct from the continuous size gradient of §4.1 (≤ 15 / 16–25 / > 25 heavy atoms), and InChIKey de-duplication across rounds prevents leakage.

The 50/50 balance is by design, and the corpus is not balanced. The sampler fills each stratum to half the round, so “98 simple / 96 complex” reflects the draw, not the literature. Under the same eligibility filters the eligible corpus is 17.5% simple and 82.5% complex (5,059 / 23,929 of 28,988; median 26 heavy atoms against the benchmark’s 20), so simple compounds are enriched 2.9 $\times$ (`scripts/corpus_reweight.py`). Balancing measures the gradient, but the headline is a weighted average under weights we chose. Reweighted to the corpus, top-1 falls from 28.4% to 15.2% [10.6–20.4] and recall from 33.5% to 19.8% [14.4–25.8] (bootstrap 95% CIs). **Verification precision has to be reweighted with them**, or the flattering half of the decomposition keeps the favourable composition: 94.3% (50/53) on simple compounds and 66.7% (8/12) on complex ones, so the pooled 89.2% becomes **71.5% [50.2–92.5]** under corpus weights — a wide interval on twelve complex compounds. We report the benchmark figure as the headline (the released set reproduces it) and the reweighted figure as the better estimate on an arbitrary paper. The reweighting assumes recovery of the raw ^1H string — the one sampler filter requiring a network fetch per record — is independent of difficulty.

Scoring is mechanical. A prediction is *correct* if its RDKit InChIKey connectivity layer (first 14 characters) matches the reference: we score *constitution*, so correct constitution with wrong stereochemistry counts as correct. The *full*, stereochemistry-sensitive InChIKey gives 21.1% top-1 and 25.8% recovered (41/194 and 50/194) against 28.4% / 33.5% — 7.3 points lower on top-1 and 7.7 on recovered — so the constitution figure is an upper bound on full-stereochemistry accuracy.

Constitution is the headline because 1D $^1\text{H}/^{13}\text{C}$ /IR rarely fixes absolute configuration: only 10.3% (20/194) of answers carry a *defined* (assigned R/S) stereocentre, and a model is penalised there for information the prompt never contained (`scripts/score_main.py --stereo`).

Metrics. *Top-1 (exact constitution)*: best-ranked candidate matches at the connectivity layer; *recovered (top-3)*: reference among up to three returned candidates, the lenient “recovery” protocol of ref.². *Generation recall*: reference in the candidate pool *before* re-ranking, the ceiling any verifier can reach (coincides with recovered except in §5.3, where the pool is larger). *Verification precision (conditional on recall)* over recall-positive compounds isolates verification from generation. The forward-verifier ranks by symmetric *chamfer distance* between predicted and observed ^{13}C peak sets (lower is better); Morgan(2, 2048) Tanimoto⁴⁰ gives a graded scaffold-family signal. CIs are bootstrap 95%; model-vs-model differences use McNemar’s exact test⁴¹ with Holm correction⁴².

Solvers are frontier LLM agents (Claude Opus), one sub-agent per batch, closed-book under a consumer subscription: no tools beyond an RDKit formula check, no ground truth, verified by grep-auditing transcripts.

Formula adherence is high but imperfect. Of the 126 candidates carried through forward-verification on the original arm (§5.2), 91.3% (115/126) match the given formula exactly, and 76.6% across the full top-3 pool. On top-1 answers adherence is 95.0% (38/40) in v3 and 90.0% (18/20) in the v2-control — the two pre-registered controlled rounds, 60 compounds between them — against 77.6% (104/134) in the headline main round. Wrong-composition answers score as misses (~one main-round compound in five), so this inflates nothing; the main-round arm is the weakest-constrained (`scripts/analyze_misses.py`).

4. How well do LLMs elucidate real structures?

4.1 Headline performance

Solver agents work blind from formula + IR + ^1H + ^{13}C , returning up to three ranked candidates in bounded contexts reset every 2–12 compounds rather than one long context — the variable §4.3 shows what matters. The benchmark comprises all 194 compounds (134 spectrally-validated + 60 controlled-round; `data/benchmark_main/raw/`).

Table 2. Headline elucidation performance on IRSpectra-Bench (n=194). Overall and by difficulty stratum, with bootstrap 95% CIs.

metric	overall (n=194)	simple (n=98)	complex (n=96)
top-1 exact constitution	28.4% [22–35]	48.0% [39–57]	8.3% [3–15]
recovered (top-3)	33.5% [27–40]	54.1% [44–63]	12.5% [6–20]
scaffold-level (best Tanimoto ≥ 0.45)	56%	73%	39%
mean best Tanimoto	0.59	0.73	0.45

A strictly-validated subset (134 main-clean + 57 controlled-clean = 191) reproduces this within ≈ 1 point on every metric — top-1 28.8%, 95% CI 23–36; recall 34.0%; simple 49.0% / complex 8.4% — so the asymmetric inclusion of pre-registered controlled sets does not drive it. The deliberate 50/50 difficulty balance does: reweighted to the 17.5%-simple composition of the eligible corpus, top-1 is 15.2% [10.6–20.4] and recall 19.8% [14.4–25.8] (§3). Per-stratum figures are unchanged; the

reweighted number is what to read as performance on an arbitrary paper. Accuracy falls monotonically with size (Fig. 2): 60.5% top-1 at ≤ 15 heavy atoms, 28.3% at 16–25, 7.0% above 25 (Fig. S3).

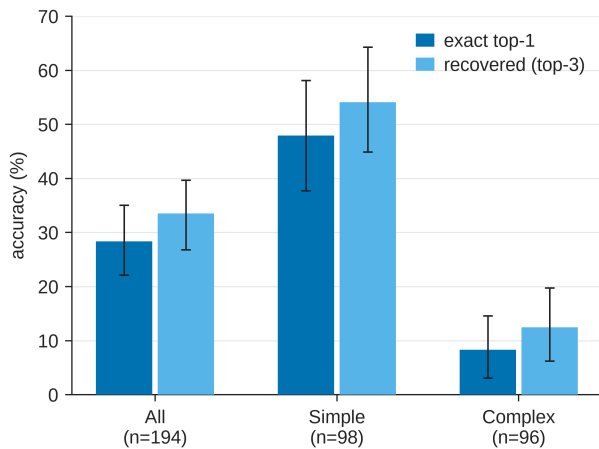


Fig. 2. Top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194), with bootstrap 95% confidence intervals. The benchmark separates a realistic difficulty range: simple targets are solved four to six times as often as complex ones on both metrics.

Of the 137 analysable top-1 misses (139 in all; two predictions did not parse; `scripts/analyze_misses.py`), 76.6% are constitutional isomers of the true structure against 23.4% with the wrong formula — not mostly regiochemistry: only 22.6% share the true Murcko scaffold, 2.9% reach Tanimoto ≥ 0.85 , and the median isomeric-miss Tanimoto is 0.39. Strict regiochemistry accounts for a fifth of failures; most misses are larger rearrangements of the same atoms. Near-degenerate $^1\text{H}/^{13}\text{C}$ shifts under-determine connectivity; the misses show *wrong connectivity at the right composition*.

4.2 Reconciling with prior reports

Our 28% top-1 sits far below the $\approx 100\%$ on “simple” molecules in a non-peer-reviewed company white paper². Read against the *peer-reviewed* record — MolPuzzle¹⁴ and re-scorings¹³, and trained baselines^{1,6} — four methodology and scoring choices differ, each raising a reported number; we name them without apportioning the gap. **Difficulty:** ring count bounds difficulty from below — our simple stratum includes a hexasubstituted benzene, while ≥ 4 rings mark 38% of recall-negative compounds against 3% of recall-positive ones (§5.3). **Scoring:** prior work counts recovery if the reference appears among three ranked candidates over three independent runs; we report single-run top-1 and top-3. **Hints:** prior hard targets got starting-material SMILES; we give none, though formula alone is worth 5% top-1 with spectra masked (§4.6). **Curation:** prior compounds were hand-selected; ours are scraped, unfiltered for solvability.

Versus trained models: a bound on the gap. No trained baseline was run on our test set; published numbers differ in spectrum realism, hints, and exact-match definition — a reason not to read the comparisons below as a ranking, but not to skip them. Running an existing open elucidator needs public weights and code accepting IR + ^1H + ^{13}C with a supplied formula; we had none. The nearest trained generator (§5.6) is ours, not independent. Most trained baselines report in-distribution accuracy on simulated spectra; Alberts et al. do not — 63.8% top-1 from IR alone, formula supplied, on experimental NIST gas-phase spectra of a curated single-instrument

library of 6–13-heavy-atom molecules¹⁸. Ours span 8–60 heavy atoms (median 20), only 15 of 194 (8%) in that window; against our ≤ 15 -heavy-atom stratum (60.5%) their 63.8% is level — level is not parity in our favour, because their model reads infrared alone where ours reads infrared and ^1H and ^{13}C . A three-modality frontier LLM matching a single-modality trained transformer does not flatter the LLM. Size alone does not explain the rest: the same work reports 59.94% over 5–35 heavy atoms ($n=5,024$), covering most of ours. The remaining difference is the spectra — one gas-phase instrument against thousands of laboratories — and the comparison bounds the gap between those regimes, not a ranking. Multimodal systems report more on data of their own making: Spectro 93% (82% with fixed embeddings) on a split whose IR is plotted from reference data and whose NMR is software-predicted¹; NMIRacle 48% top-1 / 66% top-15 on molecules held out from a *simulated* corpus inside its own training distribution⁶. Ours is 28.4% top-1 (33.5% recovered, top-3), 29.9% with forward verification (§5), on blind, real, literature-mined experimental spectra of out-of-distribution compounds.

One asymmetry cuts against us: NMIRacle takes no molecular formula where we supply it (§3), so settings differ on opposite axes — our spectra real and out-of-distribution against their simulated and in-distribution, their input strictly harder — and neither number bounds the other. As a *bound on the simulated-to-real gap*, high in-distribution accuracies substantially overstate real-world performance. Metric instability reinforces the caution: the $\approx 40\times$ MolPuzzle swing (§1.1; GPT-4o: 1.4%¹⁴ to 27.8% to 57.8%¹³, method- and harness-dependent) bounds the weight any unaudited near-100% claim² can bear.

4.3 Methodology dominates: a within-compound control

The same 20 molecules were solved two ways: (a) one LLM context, all sequential, no tools; (b) four independent agents, each on five compounds, with RDKit formula-checking. Recovered (top-3) rose from 5% to 15% ($1/20 \rightarrow 3/20$) and top-1 from 0% to 15% ($0/20 \rightarrow 3/20$). McNemar’s exact test (§3) does not reach significance — at this size it could not: the top-1 arm has three discordant pairs ($b=0$, $c=3$, $p=0.25$) and the recovered arm four ($b=1$, $c=3$, $p=0.625$). Below six discordant pairs the smallest attainable two-sided p exceeds 0.05, so “not significant” reports the design, not the data. The 15% carries a Wilson 95% CI of roughly 5–36%. Bounded, frequently-reset contexts with tool access appear to raise measured performance, consistent in direction but not established in size at $n=20$ ($p=0.25$): a directional within-compound demonstration that does not measure a $3\times$ effect. Small rounds swing widely (15–40% across $n=20$ –40 draws), hence the full 194-compound headline. This plausibly explains *part* of the gap to optimistic prior reports whose per-problem API calls implicitly used method (b) — a hypothesis, not a quantified contribution.

4.4 Model comparison: the benchmark orders capability but separates only the extremes

We solved a fixed 24-compound subset blind with four Claude models spanning a wide capability range, including the newest (Table 5), under one prompt, one scorer and one candidate budget (Fig. 3, Table 3).

One protocol asymmetry must be disclosed: the three comparison models solved the 24 compounds as four six-compound contexts each, whereas the Opus column reuses the headline run, where those items sat in one six-compound and two twelve-compound contexts — the variable that §4.3 measures at $5\% \rightarrow 15\%$. The Opus estimate is therefore not protocol-matched and, if anything, handicapped

by the longer contexts; this touches neither the nesting nor the Fable-vs-Haiku contrast, and a clean re-run is outstanding in docs/MODELS.md.

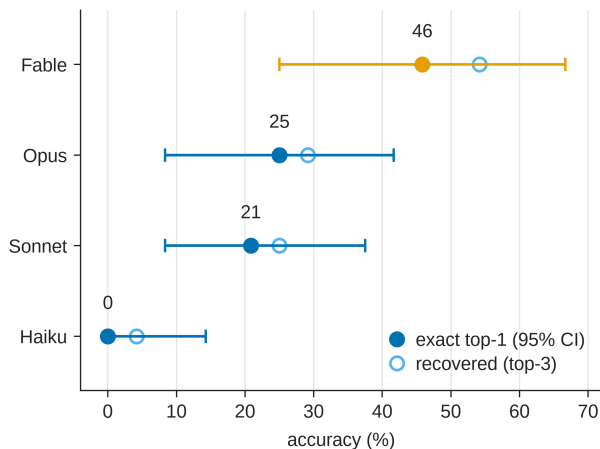


Fig. 3. Four-model comparison on a fixed 24-compound subset, solved blind under one protocol: Fable 5 46% > Opus 25% > Sonnet 21% > Haiku 0% top-1. The outcomes are strictly nested — each stronger model solves a superset of the weaker one’s compounds — so the benchmark is capability-sensitive, but at n=24 it is underpowered to separate adjacent models (§4.4).

Table 3. Four-model comparison on a fixed 24-compound subset. All four models ran the identical blind protocol.

model	top-1	recovered	top-1 95% CI
Claude Fable 5	46%	54%	[25–67]
Claude Opus	25%	29%	[8–42]
Claude Sonnet	21%	25%	[8–38]
Claude Haiku	0%	4%	[0–14]

CI’s are bootstrap except Haiku’s: Clopper–Pearson exact for 0/24, where the percentile bootstrap is degenerate.

Outcomes are strictly nested ($\text{Haiku} \subset \text{Sonnet} \subset \text{Opus} \subset \text{Fable}$), so the benchmark is capability-sensitive but at n=24 underpowered to separate adjacent models. By McNemar’s exact test only Fable-vs-Haiku survives multiple-comparison correction (Holm-adjusted $p=0.006$). Fable-vs-Opus does not: uncorrected $p=0.063$, the *floor* for its five discordant pairs, so the comparison had no way to reach 0.05 (Holm-adjusted $p=0.19$). Opus and Sonnet are indistinguishable. Two mid-tier models agreeing closely puts the recall-bound regime of §4.1 beyond a single model; the newest nearly doubles the next-best top-1 yet still misses the majority. We used Claude-family models callable under one subscription; the monotonic spread makes lineage-specificity unlikely.

4.5 Domain case study: battery-electrolyte chemistry

To test the recall-bound regime on battery-electrolyte chemistry rather than on organic molecules at large, we curated *IRSpectra-Bench-Electrolyte*: a 48-compound subset of IRExp records across the six functional families that dominate lithium- and sodium-battery electrolytes, eight compounds

per class, held out of every other split and solved blind under the identical protocol. Two yielded no parseable candidate, leaving 46 scored. These are literature compounds, not operando or in-cell spectra, and we make no claim that they are.

Performance matches the headline regime — top-1 26%, recovered 28% (12/46 and 13/46; at $n=46$ the 95% CI is wide and overlaps the headline), scored over the 46 parseable compounds — consistent with a task bottleneck, not a chemical neighbourhood. The main round scores an unparseable answer as a miss, so this arm also reads $12/48 = 25\%$; we report both denominators. The true structure enters the candidate set for only 13 of 46 compounds and ranks first in 12 of those 13. Forward-verification was not run here, so this is the recall-bound pattern under the solver’s own ranking rather than the §5 measurement (Fig. S4). The per-class breakdown is Table 4; at eight compounds per class the intervals overlap throughout.

Table 4. Per-class performance on IRSpectra-Bench-Electrolyte ($n=46$).

electrolyte class	n	top-1	95% CI	recovered (top-3)	95% CI
sp ³ -C-F	8	50%	[22–78]	50%	[22–78]
carbonate	7	29%	[8–64]	43%	[16–75]
phosphoryl	8	25%	[7–59]	25%	[7–59]
glyme / oligoether	7	29%	[8–64]	29%	[8–64]
sulfonyl / sulfonate	8	12%	[2–47]	12%	[2–47]
nitrile	8	12%	[2–47]	12%	[2–47]

Intervals overlap throughout (six-class χ^2 $p \approx 0.56$; best-vs-worst Fisher exact $p \approx 0.28$), so the ordering is hypothesis-generating, not established — C-F couplings pin sp³-C-F regiochemistry, while sulfonyl and nitrile targets turn on oxidation-state ambiguity and heteroaromatic substitution that ¹H/¹³C shifts underdetermine. It is the recall-bound failure of §4.1 in a domain-specific guise, and it is where §5’s *forward-verification* recipe plays a role *analogous* to (not a replacement for) computational-NMR validation.

4.6 Is the model reading the spectra? A formula-only control

Every benchmark compound is mined from open-access literature, so a frontier model may have met it in pretraining, a well-documented hazard for LLM evaluation⁴³. We reran the identical blind protocol with every spectral channel masked, leaving the solver the molecular formula and nothing else. A formula does not determine constitution, so accuracy materially above the floor would indicate recall rather than reasoning.

The arms were not generated together: the *formula-only* arm was fresh (2026-07-28), the *full-modality* arm archived from June. The bias therefore runs *toward* formula-only — fresh agents reason harder per compound than the archived batched run — and it still collapsed to 5%; a confound that works against the finding cannot manufacture it.

Table 5. Formula-only control. Paired on the same 60 compounds as §5.

condition	top-1	recovered (top-3)
formula only	3/60 (5%)	3/60 (5%)
formula + IR + ^1H + ^{13}C	14/60 (23%)	19/60 (32%)

The outcomes are perfectly nested: eleven compounds are solved with the spectra and not without, and none the other way round (McNemar exact $p=0.001$). Removing the spectra removes the result.

A second, independent control: publication recency. The formula-only arm shows the spectra carry the signal, not whether memorisation contributes to the part they explain; if recall drove the headline number, accuracy should fall with recency. Across all 194 compounds, source papers spanning 2008–2026, accuracy is flat: 28.6% for the older half (≤ 2020 , $n=112$) against 28.0% for the newer ($n=82$), point-biserial $r=-0.007$, and no monotone trend across buckets (Fig. 4b), the most recent (≥ 2024 , $n=25$) highest at 40% [23–59].

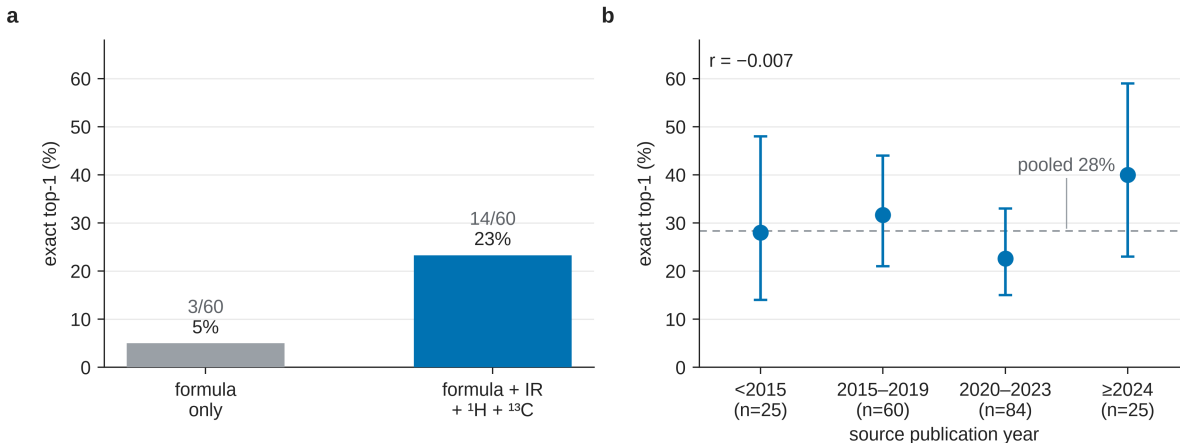


Fig. 4. Two contamination controls. (a) The solver reaches 3/60 on formula alone against 14/60 with IR + ^1H + ^{13}C , nested outcomes: 11 solved only with the spectra, none only without (McNemar exact $p=0.001$). (b) Accuracy against source-paper year ($n=194$, Wilson 95% CIs) is flat, point-biserial $r=-0.007$, where pretraining recall predicts a decline.

Newer papers skew larger (median 22 heavy atoms against 20) and size dominates accuracy, biasing the raw split *against* them (§4.1); size-adjusted, the older-minus-newer difference is -5.1 points, 95% CI $[-17.2$ to $+7.0]$ (Cochran–Mantel–Haenszel $\chi^2=0.42$, $p=0.51$, continuity-corrected). The sign opposes pretraining recall, but the interval includes zero, so this bounds any recency effect; by the standard that §5.3 applies to adjacent conditions, we do not read it as directional.

The 5% is not zero, and its three compounds do not all read the same way: one is absent from PubChem and near-determined by its benzyne-precursor composition, while a catalogued sulfonamide and a named natural product are as consistent with recognition as with inference. Formula-level recall accounts for at most about a fifth of the headline accuracy on this set. Both controls agree but neither is randomised — year is observational, and formula-only cannot distinguish memorisation from inference on a near-determining formula — so we claim a strong bound rather than exclusion (§8). Neither addresses retrieval on the spectral string the prompt carries verbatim; §8 names the control that would settle it.

4.7 Does the diagnosis hold outside one vendor? A four-vendor replication

Every number so far comes from one lineage, the external-validity gap §8 (iii) named as most important. Three terms carry the table below, and §5 treats them in full. *Forward-verification* re-ranks a solver’s candidate structures by predicting each one’s ^{13}C spectrum and matching it against the observed spectrum. *Generation recall* is the fraction of compounds whose true structure the solver proposed at all, and *verification precision (conditional on recall)* is the fraction of those recalled compounds that the re-ranking then puts first. The *60-compound arm* is the two pre-registered controlled rounds, v3 and v2-control, on which the original decomposition was run (Table 7’s first column). One precision artefact: a single-candidate compound is scored correct by any re-ranker, so singleton-rich sets inflate precision by construction — hence the last column, restricted to compounds where more than one candidate existed.

Six non-Claude models solved the same 60 compounds as Table 7’s first column under the identical blind protocol; the three whose output was well-formed enough were carried through forward-verification (Table 6).

Table 6. Cross-vendor decomposition on the 60-compound arm. Recall and precision have different denominators, so the criterion is the inequality rather than a difference.

model	generation recall	verification precision recall	multi-candidate only
Claude Opus (Table 7)	19/60 = 32% [21–44]	16/19 = 84% [62–94]	10/13 = 77%
Grok 4.6	32/60 = 53% [41–65]	20/32 = 62% [45–77]	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50% [38–62]	22/30 = 73% [56–86]	22/30 = 73%
GPT-5.6 Sol	25/60 = 42% [30–54]	17/25 = 68% [48–83]	16/24 = 67%

Verification precision exceeds generation recall in every arm (Fig. 5), across four independent model families. The claim rests on the paired within-vendor difference — verification precision conditional on recall minus generation recall — not on marginal interval overlap. Resampling compounds (`scripts/cross_vendor_gap.py`) separates it from zero in three arms: Claude +52.5 points [+31.2 to +71.7], GPT-5.6 Sol +26.3 [+4.5 to +48.6] and Gemini +23.3 [+3.2 to +44.3]. Grok does not — +9.2 [−11.7 to +30.0] — and we report its gap as directional. Claude’s is the widest and least informative: singleton composition inflates it.

Six of Claude’s nineteen recall-positive compounds carried one candidate, so any verifier scores them by construction (§5.2); here that inflates Claude’s precision alone. Newer models almost always return three candidates (0–1 singletons), and on multi-candidate sets the four precisions span 62–77% while recall spans 32–53%: the limiting factor is the generator. Three models beat ours on generation recall — the diagnosis working, recall being the movable factor.

The candidate budget is not matched, and the correction runs both ways. Ours holds 2.20 candidates per compound against exactly 3.00 for every comparison model (Composer 2.82); a longer list can only raise recall, tilting the comparison toward other vendors — the mirror of the singleton correction on precision. At one candidate, the only budget every model met, recall is 14/60 = 23% for ours against 23/60 = 38% for Grok, 23/60 = 38% for Gemini and 21/60 = 35% for GPT-5.6 Sol (`scripts/cross_vendor_budget.py`). Three models still lead ours, but Grok’s margin is 15 points. The 21-point figure is not one a budget-matched comparison supports.

Three weaker models are excluded from the decomposition: Composer 2.5 (20% recall) and GPT-5.6 Luna (15%) match the given formula only 67% and 76% of the time, and `nvidia/nemotron-3.5-lightning` managed 2%. §4.1’s scorer now prints formula adherence beside recall. A seventh arm, DeepSeek V4 Pro, returned answers for only 18 of 60 compounds before the run ended; the true structure appears for 8 of those, so its recall is a lower bound and it is excluded from the comparison rather than from the release — both its solve and verify files are deposited.

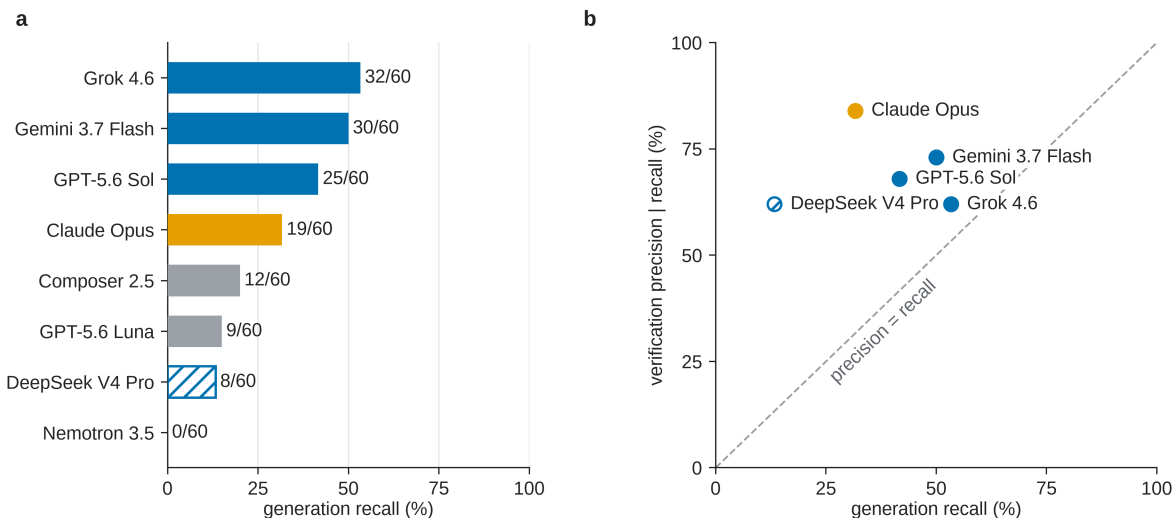


Fig. 5. Every model measured on the 60-compound arm. **(a)** Generation recall, each model at the candidate budget it used: 3.00 candidates per compound for all but ours (2.20) and Composer (2.82). That axis therefore favours the models that always returned three. Grey marks models below the formula-adherence floor, whose recall is not a chemistry result, and hatching marks an incomplete arm — DeepSeek V4 Pro answered 18 of 60 — whose recall is a lower bound. **(b)** The claim itself: verification precision against generation recall, with the diagonal. Every model sits above the line, so verification is better than generation for all four families; the hatched hollow marker is again the incomplete arm. The four Claude models of §4.4 are deliberately absent: they ran a different 24-compound subset, and putting them here would imply a comparison that was never made.

Contamination was controlled. These runs used cloud agents with repository access to tracked answer keys. Grok re-solved all ten batches from a clone with answer files removed: recall 28/60 against 32/60, paired McNemar $p=0.39$, formula adherence 97% against 95% of candidates (`scripts/cross_vendor_sweep.py`). A key-reader would solve a *superset*, yet four compounds were solved only without the key, with 24 of 60 solved by both — sampling noise, not copying. On constitutionally correct structures whose reference carries assigned stereocentres, which 1D spectra cannot supply, Grok reproduced 0/3 correct descriptors, Gemini 0/2 and GPT-5.6 Sol 0/2.

Two limits. Models were reached through a coding-assistant harness exposing reasoning-effort tiers, a decoding control the Claude arm never had (§9) — these measure a model *as served*, not a bare endpoint, and which tier served each arm was not recorded. The allowlist carries both `gpt-5.6-sol-xhigh` and `gpt-5.6-sol-high`, and the run did not report its selection, so arms are not demonstrably matched. That bounds what this section can claim unevenly: within-arm recall and precision share one run at one tier, so the inequality is internally valid; a recall *ranking between named models* orders them at unknown effort, and the 15-point margin above should be read as such. The clean-clone control covers only Grok; Gemini and GPT-5.6 Sol rest on the stereochemistry argument and the shared protocol.

5. Forward-verification elucidation

5.1 Method

The inverse direction is the model’s hard, isomer-blind direction; the forward direction (structure $\rightarrow$ spectrum) is its easy, accurate one². We close a training-free generator–verifier loop:

generate candidate structures (inverse) $\rightarrow$ *forward-predict* each candidate’s ^{13}C spectrum, blind to the observed spectrum $\rightarrow$ *re-rank* candidates by the distance between predicted and observed $^{13}\text{C} \rightarrow$ return the best match.

The inverse solver’s 373 deduplicated candidates across 194 targets are forward-predicted in shuffled, anonymised batches, blind to the observed spectrum and target identity. Predicted and observed ^{13}C peak sets are compared by symmetric chamfer distance (Fig. 6).

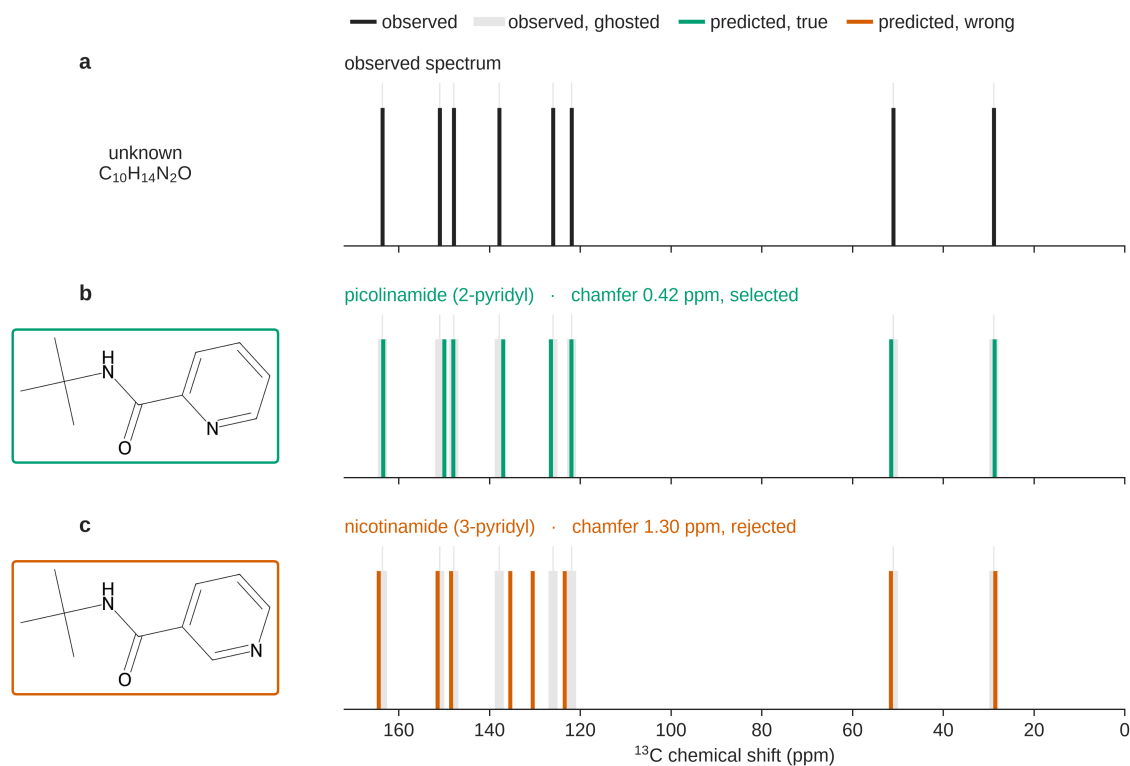


Fig. 6. Forward-verification on a benchmark regioisomer pair: picolinamide and nicotinamide are indistinguishable to the inverse task, but the true isomer’s forward-predicted ^{13}C spectrum matches the observed one at a chamfer of 0.42 ppm against 1.30 ppm for the alternative (§5.1).

Regioisomers have *different* forward-predicted shifts, but the margin is thin: isomeric candidate pairs for the same target separate by a median 1.21 ppm (quartiles 0.84 and 1.78), and 82% are predicted closer together than the predictor’s own ≈ 2 ppm error — usually within the noise. The ranking nevertheless recovers the right structure when it is present (§5.2), a real effect on a thin margin, confirmed against a derangement null — every candidate set re-scored against another compound’s observed spectrum so that none keeps its own — at $p=0.001$ (§5.5).

Verification precision runs 72–89% rather than 100%: near-degenerate candidates the predictor cannot resolve are its false positives. That spacing also sets the 54.0% derangement chance floor on multi-candidate compounds (§5.5) and the ceiling neither the HOSE lookup nor the GNN escapes (§5.4) — limits that §5 reports separately. Sharper predictors would move them, not better search; ≈ 1 ppm message-passing ensembles⁴⁴, a DFT-coupled graph network at 0.94 ppm⁴⁵, a community benchmark⁴⁶ and curated shift sets⁴⁷ exist for calibration.

The scheme is an analogy to DP4^{21,22} and NMR crystallography^{23,24}, not an equivalence: we forgo DFT’s ≈ 1 –2 ppm accuracy and its calibrated, probabilistic error model for a predictor that runs in seconds.

5.2 Result

The first column below is the 60-compound arm (v3 + v2-control) on which §5.3–§5.5 build; the second, the full benchmark. All 373 candidates were forward-predicted — nothing here is a lower bound; the self-ranking row re-derives Table 2 from §4’s output.

Table 7. Forward-verification decomposition. Original arm and full benchmark.

	60-compound arm	full benchmark (n=194)
generation recall	19/60 (32%)	65/194 (34%)
top-1, solver self-ranking	14/60 (23%)	55/194 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58/194 (30%)
<i>conditional on recall</i> — self-ranking	14/19 (74%)	55/65 (85%)
<i>conditional on recall</i> — forward-verification	16/19 (84%)	58/65 (89%)
...single-candidate compounds within that set	6/19	28/65
<i>multi-candidate only</i> — self-ranking	8/13 (62%)	27/37 (73%)
<i>multi-candidate only</i> — forward-verification	10/13 (77%)	30/37 (81%)

When the true structure is among the candidates, forward-verification selects it in 58 of 65 cases (89%). Of the 65, 28 had a single candidate — scored by construction by any ranker — so the pooled figure is not the verifier’s margin over chance. Where a choice existed (n=37), forward-verification gets 30/37 (81%) against a 54.0% derangement floor (§5.5) — +27.1 points; self-ranking 27/37 (73%).

The margin over self-ranking remains small and unresolved: seven compounds gained, four lost, McNemar exact p=0.55. We do not claim it. Precision is high in absolute terms while recall binds: top-1 moves only 28% $\rightarrow$ 30% because the true structure was never proposed for 129 of 194 compounds, which no re-ranking can repair. The verifier converts recall almost completely on *simple* targets (50/53, 94%) and ties self-ranking on *complex* ones (8/12, 67%). Elucidation factorises into two near-independent levers: a strong verifier and a generator at 34% recall that is the wall (Fig. 1).

5.3 Generate-wide: testing the recipe

The decomposition implies *generate wide, verify by forward prediction* — a chemistry analog of self-consistency sampling⁴⁸. Ten solver agents proposed up to six regiochemistry-aware candidates per compound, pooled with the originals and re-ranked.

Table 8. Generate-wide vs original. On the 60-compound arm only, since wide generation was run there; the “original” column is Table 7’s first.

	original (60-compound arm)	generate-wide
generation recall (true structure among candidates)	32%	42%
forward-verified top-1	27%	30%
verification precision (conditional on recall)	84%	72%

Wide generation lifts recall +10 points (32% → 42%, 19/60 → 25/60) and top-1 +7 over self-ranking (23% → 30%, 14/60 → 18/60), i.e. +3 over forward-verified top-1 (27% → 30%, 16/60 → 18/60, Table 8), on the same 60 compounds (Fig. 7) with no training.

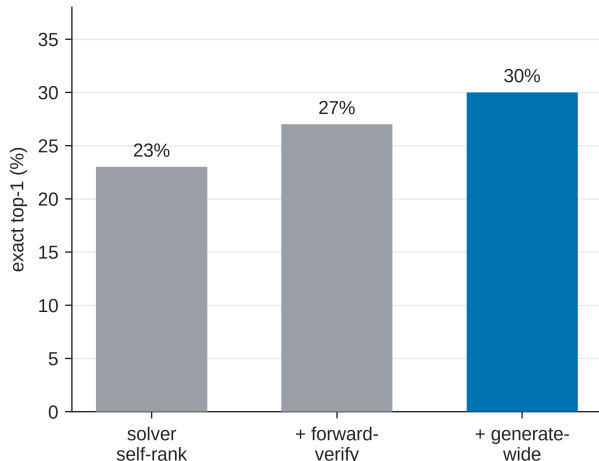


Fig. 7. The forward-verification inference ladder on the 60-compound arm: solver self-ranking → + forward-verification → + generate-wide, at 23% / 27% / 30% top-1. Each rung is a training-free change to inference alone, directional rather than individually resolved at this sample size (§5.3).

These top-1 differences are directional; at n=60 they are not statistically resolved. Self-ranking to generate-wide gains seven and loses three, McNemar exact p=0.34 (b=3, c=7). The recall gain is better supported: the ladder shows the *mechanism* works and recall is the movable factor, not how large the top-1 gain is.

The ladder does not extend. (i) Recall plateaus at 42%, on large polycyclic targets rather than exotic ones — ≥ 4 rings in 3.1% of recalled compounds against 38.0% of missed ones. (ii) Verification precision falls 84% → 72% as more near-degenerate regioisomers enter the pool. This recall/precision tension is the ceiling of the training-free approach: closing the gap requires sharper verification or 2D-NMR constraints, not merely more candidates.

How much of the wall is regiochemistry? Enough to matter and not enough to remove it. Of the 129 compounds the solver misses, 97 (75.2%) have a candidate of the *right molecular formula* and 41 (31.8%) share both formula and generic scaffold — the class a deterministic isomer enumerator can reach (`scripts/analyze_recall_headroom.py`). Adding those to the 65 already recalled gives a loose upper bound of $106/194 = 54.6\%$; realised reach is 41.8% (§5.3). Roughly a third of the remaining wall is substituent placement around a scaffold the model already has; two thirds, a scaffold it never proposed. Generic-scaffold equality overcounts what an enumerator can build; we report the realised figure as the achievable one.

5.4 Non-LLM verifiers: a deterministic lookup and a learned model

§5.3 suggests replacing the verifier with a non-LLM ^{13}C predictor. We tested a HOSE-code⁴⁹-style lookup and a small message-passing GNN, both trained on the same nmrshiftdb2 dump⁵⁰ and applied to the same §5.2 candidate sets so only the predictor changes. The GNN is deliberately modest; sharper purpose-built ^{13}C models exist^{44–46}.

Table 9. Verifier comparison, conditional on recall.

verifier	60-compound arm (n=19)	full benchmark (n=65)	held-out ^{13}C MAE
solver self-ranking	14/19 (74%)	55/65 (85%)	—
deterministic HOSE <i>lookup</i>	14/19 (74%)	55/65 (85%)	3.23 ppm
learned GNN (same data)	16/19 (84%)	59/65 (91%)	1.70 ppm
LLM forward-verifier (§5.2)	16/19 (84%)	58/65 (89%)	—

The lookup ties the solver’s own score at both sample sizes (Table 9). Coverage explains why: of the 6,360 candidate carbons, only 2% match a training environment at $r=4$ and 71% resolve at $r\leq 2$ or the hybridisation prior — too coarse to separate regioisomers.

The GNN matches and slightly exceeds the LLM verifier, 59/65 against 58/65 (Fig. S6), though not significantly: *suggestive and directional*, consistent with the lookup’s failure being substantially method rather than coverage alone, but not established. Neither predictor reaches the near-degenerate-regioisomer precision ceiling (§5.3/§5.5), where DFT-level accuracy or orthogonal 2D-NMR constraints remain the genuine fix.

Leakage is slight: of the 373 candidate structures the two predictors re-rank (364 distinct InChIKey-14), two appear in nmrshiftdb2, both distractors on recall-negative compounds; none of the 65 benchmark-answer candidates appears there (`scripts/verifier_leakage.py`). Y-randomisation (1,000 derangements) places the real result above the 97.5th percentile (n=19: real 84% vs mean 58.6%, one-sided $p<0.05$). Like §5.6, the learned verifier is a trained complement, reported outside the training-free protocol.

Concurrent industry work adds a fourth comparator outside our protocol. Wagen gave a frontier LLM up to ten calls to MagNET, an equivariant graph transformer whose predictions on true structures reach 1.20 ppm MAE and 1.60 ppm RMSD against experiment⁵¹, on eight ^{13}C problems from structures misassigned and later corrected computationally⁵², at 56 runs per configuration⁵³. Scored as we score on connectivity, the external predictor moves 46.4% (26/56) to 55.4% (31/56) — the same order as our ladder (§5.3); the larger 21.4% to 51.8% headline is exact SMILES including

stereochemistry, which our protocol does not test (§8). The author reports the study as underpowered and preliminary at $n=8$ and flags possible training-set leakage, so we read it as evidence of direction rather than magnitude.

5.5 Negative control

Permutation negative control (Y-randomisation analog). A re-ranker exploiting genuine predicted-vs-observed ^{13}C agreement must lose it when the pairing is broken. We re-paired each candidate set with a deranged observed ^{13}C spectrum — no compound keeps its own — and re-ran the verifier 1,000 times. Verification precision falls from 89.2% (58/65) to a permuted mean of 73.8% (95% range 66.2–81.5%; one-sided $p=0.001$, two-sided $p=0.002$). The verifier acts on real spectral agreement, not a candidate-list artefact. That floor is high because 28 of the 65 recall-positive compounds carry a single scorable candidate, scored correct under every pairing and contributing no verification signal. On the 37 that carried more than one candidate — §5.2’s verification set — the floor falls to 54.0% and the real value is 81.1% (30/37), a margin of +27.1 points rather than +15.4, at the same one-sided $p=0.001$. We report the restricted figure as the verifier’s margin over chance and the pooled one for continuity with §5.2’s denominator.

Confidence calibration (a negative result). Ranking the 138 multi-candidate compounds of the whole benchmark — more than one candidate, of the 192 that received any, wider than the 37 recall-positive set above — by chamfer margin and answering only the most-confident fraction leaves top-1 flat and non-monotonic (22% at full coverage, 24% at 75%, 28% at 50%, 24% at 25%). Single-candidate compounds, which have no margin, must be excluded; retaining them produced a spurious “improvement”. We do not claim the verifier distance as a calibrated abstention gauge (§5.4). This does not contradict chamfer distance being informative in aggregate (§5.2): the *absolute* distance separates right from wrong across compounds, while the *within-compound margin* does not rank one compound’s confidence against another’s. Selecting which questions to answer needs the second; only the first is established here.

5.6 Is the recall wall task-intrinsic? A trained-generator probe

That ceiling is a property of *training-free LLM elicitation*; whether the $\approx 42\%$ recall plateau is intrinsic to 1D-data elicitation is separate. We probed it by substituting a small ($\approx 16\text{M}$ -parameter) $^1\text{H}/^{13}\text{C} \rightarrow \text{SMILES}$ transformer for the enumerator, pooling its candidates with Claude’s and re-ranking with the same verifiers.

On the 194-compound benchmark the true structure enters the pool for 54.1% of compounds, versus 41.8% for scaffold enumeration and 33.5% for Claude alone. Neither of the first two arms is an independent elucidator: enumeration relocates substituents *on a model candidate* (`scripts/enumerate_isomers.py`) and is Claude-seeded; the generator’s candidates are pooled with Claude’s. All three rows measure what is added to the LLM’s pool, not what a system reaches alone. Where enumeration’s near-degenerate regioisomers *collapse* the HOSE verifier (top-1 28.4% $\rightarrow$ 16.0%, the §5.3 precision-loss mechanism), the generator’s formula-correct, ^{13}C -separable candidates convert: top-1 rises 28.4% $\rightarrow$ 35.1% (McNemar exact $p=0.015$; +6.7 points, 95% CI [+2.1 to +11.9]; Fig. S5). Corrected over the probe’s three arms — Claude alone, plus enumeration, plus generator — it survives Holm (0.029) and Benjamini–Hochberg (0.022); it would not survive correction over every p -value in the paper, and we name the family rather than leave scope to be guessed. With the LLM forward-verifier on the 60 forward-verify compounds, recall rises 42% $\rightarrow$ 57% (34/60) and top-1 reaches 47% (28/60), at 82% verification precision (28/34). These figures come from a re-run, not

a reproduction: an earlier pass reported 41% top-1 at 73% precision but deposited no outputs; the re-run lands above it.

Two controls guard against memorisation: the fine-tuning split removes every benchmark InChIKey-14 ($\text{train} \cap \text{benchmark} = 0$, $\text{val} \cap \text{benchmark} = 0$), with none of the 40 newly-recovered compounds in either stage (0/40 in simulated pretraining, 0/40 in fine-tuning); and the simulated-pretrained model alone recovers 0/248 zero-shot on the probe’s held-out split — all four cohorts pooled and de-duplicated at InChIKey-14 (140 main, 48 electrolyte, 40 v3, 20 v2-control), a superset of the 194-compound benchmark — rising to 25% on that same 248-structure split only after IReXP fine-tuning (`contrib/generator_probe/`). IReXP data, not architecture, is the active ingredient; the recall ceiling is elicitation-specific rather than task-intrinsic. We report this as a probe, not part of the headline training-free protocol.

Purpose-trained systems make the point more strongly: NMR-Solver reaches 52.9% top-1 on experimental literature $^1\text{H}/^{13}\text{C}$ with formula supplied²⁷, and a purpose-trained IR transformer reaches 63.8% top-1 on experimental NIST spectra¹⁸. Whatever bounds a training-free LLM at 28–30%, it is plainly not the task. The public split `irexp_release/train` does *not* hold out the benchmark (117/200 InChIKey-14 overlap), so downstream users must de-leak; see *Data availability*.

6. The decomposition across the published literature

The split measured above needs no new data to apply to published work. A system reporting top-1 = a and top- k = b has already said: the true structure was in its candidate set at least as often as it reached top k , so **recall** $\geq b$; conditional precision is $\leq a/b$, since any recall above b divides the same a by a larger denominator; and at least $b - a$ compounds were proposed then mis-ranked. Where the emitted list is capped at reported k , the floor is exact — Alberts et al. generate ten ranked SMILES per sample — and where the internal pool is larger it is a bound, as with NMR-Solver’s documented 1,000-molecule pool behind a top-10 column. Table 10 applies this to every system we could read in full (`scripts/literature_decomposition.py`).

One caution governs the table. The a/b ceiling bounds ranking a pipeline *already does*, not what any verifier could achieve: our solver ranks $55/65 = 84.6\%$, recoverable as $28.4/33.5 = 84.8\%$ from published figures alone; forward-verification reaches $58/65 = 89.2\%$ by re-ordering. The gap from a/b to 100% is the room a re-ranker has.

Table 10. Recall and ranking loss from published top- k figures. Rows are comparable only within a correctness criterion; stereochemistry-strict and connectivity figures differ by 21 points on NMR-Solver’s identical predictions. Candidate budgets are given because recall over three candidates is mechanically smaller than over ten.

system	data (n)	top-1	top- k (k)	recall	prec.
<i>scored on connectivity</i>					
this work, solver alone	literature (194)	28.4%	33.5% (3)	33.5%	84.6%
this work, + forward-verification	literature (194)	29.9%	33.5% (3)	33.5%	89.2%
NMR-Solver ²⁷	literature (450)	52.9%	67.3% (10)	$\geq 67.3\%$	$\leq 78.6\%$
NMRAgent ²⁸	literature (450)	61.6%	70.0% (10)	$\geq 70.0\%$	$\leq 88.0\%$
<i>scored with stereochemistry</i>					
this work	literature (194)	21.1%	25.8% (3)	$\geq 25.8\%$	$\leq 81.8\%$
NMR-Solver ²⁷	simulated (1,000)	66.9%	89.9% (10)	$\geq 89.9\%$	$\leq 74.4\%$
NMR-Solver ²⁷	literature (450)	31.6%	53.8% (10)	$\geq 53.8\%$	$\leq 58.7\%$
Espejo Morales ³⁰	education (236)	80.9%	90.0% (5)	$\geq 90.0\%$	$\leq 89.9\%$
Espejo Morales ³⁰	industrial (34)	20.6%	29.1% (5)	$\geq 29.1\%$	$\leq 70.9\%$
<i>exact match, stereo handling not stated</i>					
Alberts, 6–13 atoms ¹⁸	NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
Alberts, 5–35 atoms ¹⁸	NIST IR (5,024)	59.9%	78.5% (10)	78.5%	76.4%
SpecX, random split ²⁰	simulated (99,439)	59.0%	81.8% (10)	81.8%	72.2%
SpecX, scaffold split ²⁰	simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%
IR-Agent ¹⁵	NIST IR (905)	10.3%	21.6% (10)	21.6%	47.7%
<i>criterion not stated</i>					
Priessner ¹⁶	experimental (34)	20.6%	—	26.5%	77.8%

Three groups have already run the control this paper needed. Hold system, criterion and candidate budget fixed; change only the data. Since top-1 = recall $\times$ precision, each term’s share of the collapse is its share of the log ratio — the absolute gap $b - a$ would mislead, bounded by b and shrinking whenever recall does. NMR-Solver moves from simulated to real literature spectra and loses 35 points of top-1, of which **recall carries 68%**. SpecX’s transformer moves from random to scaffold split: **70%**. Espejo Morales et al.’s agent moves from chemistry-education

to industrial samples: **83%**. Ranking degrades in all three — relative loss 25.6% $\rightarrow$ 41.3%, 27.8% $\rightarrow$ 41.3%, 10.1% $\rightarrow$ 29.1% — so verification is not unaffected. The claim is which term dominates, and in every pair it is the same one. None of the three papers reports this or was testing it.

Published gains have come almost entirely from the supply side. NMRAgent’s ablation runs 36.89 $\rightarrow$ 60.47 $\rightarrow$ 64.27 $\rightarrow$ 64.42: retrieval and de-novo generation account for 27.38 of the 27.53-point span, and the verifier-driven optimisation loop that earns the paper *evidential* is worth 0.15 points²⁸. NMRGym’s formula ablation moves recall at $k=10$ from 17.37% to 36.48% while conditional precision moves only 36.1% $\rightarrow$ 49.1% — the constraint buys candidates, not judgment¹⁹. IR-Agent’s agentic layer adds 4.0 points of recall against 0.5 of top-1¹⁵.

The regime has a boundary. Since $-\log(\text{top-1}) = -\log(\text{recall}) - \log(\text{precision})$, recall’s share of a system’s total loss is $\log(\text{recall})/\log(\text{top-1})$ — scale-free, and unlike a miss-to-ranking ratio not driven by where the reported k happens to fall. Among rows where recall is exact that share is 38% on SpecX’s simulated random split and 39% on Alberts’ NIST gas-phase library, where ranking is therefore the larger term; 56% on SpecX’s scaffold split; 67% for IR-Agent on experimental NIST spectra; and 87–91% on ours. What separates the low end from the high is not method sophistication — both extremes are trained transformers — but how far evaluation data are from one curated distribution.

Candidate budget confounds this and pushes it our way: recall over three candidates is mechanically smaller than over ten, which raises our share. Our generate-wide arm supplies the test, widening the budget on the same compounds — at recall 42% and top-1 30% the share is 72%, down from 91% but still far above the 38–39% of the simulated and single-library rows (§5.3). The ordering survives the confound; the magnitude should be read with it in mind. Ranking is co-equal or dominant where spectra are simulated or drawn from one curated instrument library with the formula supplied; recall binds where they are real and heterogeneous. The clearest counterexample is Priessner et al., whose reasoning-LLM re-ranker moves top-1 from 41.2% to 67.6% on a fixed candidate pool¹⁶ — 26 points bought purely by re-ordering. That is their condition in which the true structure seeds the pool, which is not the row tabulated above and is why its recall is high by construction; the tabulated row is the harder, analogue-seeded one.

One number the table explains with our own measurement. NMR-Solver’s conditional precision is bounded at 78.6% — below our pooled 89.2%, though above the 71.5% we get under corpus weights, and we say so because §3 calls the reweighted figure the better estimate. Either way the ordering is not what a sharper predictor predicts: theirs is roughly twice as sharp as ours. The explanation is pool width rather than predictor sharpness, and we measured it directly: candidates for one target are predicted a median 1.21 ppm apart, inside the predictor’s own error for 82.1% of isomeric pairs (`scripts/isomer_separability.py`), so every candidate added to a pool adds a near-degenerate competitor. Our ladder shows the same tension from the inside — verification precision falls from 84% to 72% as the pool widens (§5.3) — and NMR-Solver ranks a documented 1,000-molecule pool where we rank three. Conditional precision and recall trade against each other along the candidate budget, which is why neither number means much without the other, and why we report the budget with both.

What the field does not measure. Across every paper we read, the frequency with which the true structure was proposed at all appears once, in a supplementary figure caption¹⁶. Espejo Morales et al. declare a ten-candidate output and report $k = 1, 2$ and 5³⁰; NMRAgent’s verifier emits a `need_bigger_pool` verdict and the paper never reports how often it fires²⁸. One collision is worth naming: NMR-Solver defines *recall@K* as the proportion of cases where the correct structure

appears among the top K candidates²⁷, so its “52.89% recall” is a top-1 accuracy and is not the quantity we call recall. Setting our 33.5% beside it would compare incommensurable numbers.

7. Discussion

The Claude-family models we tested are reliable scaffold-level elucidators and good *verifiers* (verification precision 89%); exact top-1 is throttled by generation recall and by the regiochemical underdetermination intrinsic to 1D NMR. The contribution is a diagnosis with a bounded, training-free improvement attached, not a method that solves the task.

The diagnosis held under every perturbation we applied — four Claude models, a second chemical domain, four verifiers — and reproduces in one application domain (§4.5): *generation recall* binds, consistent with a structural bottleneck. The same lens on the literature (§6) locates the boundary: recall binds where spectra are real and heterogeneous; ranking dominates where they are simulated or from one curated library. Among systems whose recall can be read from their own figures, recall’s share of total loss runs 38–39% (simulated split, NIST gas-phase library) to 87–91% (ours). Three groups changed only their data and recall carried 68%, 70% and 83% of each collapse. Forward-verification moves top-1 from 28% to 30% (§5.2); recall plateaus at 42% (§5.3).

This reframes the engineering problem away from “do not train a model”. NMR-Solver reaches 52.9%, a trained IR transformer reaches 63.8%, against our training-free 28–30%; our 16M-parameter generator lifts pooled recall to 54.1% and top-1 to 35.1% (§5.6). The claim is narrower and survives that. What ages out is a *particular* trained artefact; what compounds is open experimental data and an honest benchmark. The same architecture on a 248-structure held-out split recovers 0/248 without IReXP and 25% with it — the released data, not the architecture, is the active ingredient. Inference-time scaffolding is the second durable lever: it needs no training and rides each new model.

Concurrent work sharpens both halves. Kliuev *et al.* independently place the best of six frontier LLMs at 24% on 105 experimental $^1\text{H}/^{13}\text{C}$ peak lists²⁹, within a few points of our 28%, so the headline is not an artefact of our corpus or our model. Espejo Morales *et al.* then push the lever our decomposition names as binding harder than we do: environment design alone — processing tools, tabulated shifts, hard constraint checks, and no shift-matching ranker — carries a frozen LLM to 71.1% top-1 on the 15-molecule set where graduate students reach 66.7%, against 24.4–40.0% for frontier LLMs given the same inputs in a bare coding harness³⁰. That movement is entirely on the proposal side, where we locate the headroom; what it does not report is the split that locates it. And their own 80.9% on curated education spectra against 20.6% on 34 industrial samples is our central empirical claim, measured by someone else. Wagen’s MagNET-in-the-loop study is the complementary lever: an external 1.20 ppm ^{13}C predictor lifts exact-SMILES success from 21.4% to 51.8% on eight ^{13}C -only problems, while on connectivity the move is 46.4% to 55.4% — the same order as our ladder, and the larger jump is on stereochemistry we do not score⁵³. Espejo’s environment, Wagen’s simulator and our generate-wide arm therefore attack three stages of one inverse problem: proposal, ranking, and the candidate budget that couples them.

Protocol is a lever of the same order as capability: a number quoted without its protocol is uninterpretable. We do *not* claim protocol dominates capability — the model axis spans 0–46% top-1 on the fixed 24-compound subset (Table 3), wider than the 5% → 15% protocol effect (§4.3), on different sets. Learned molecular embeddings rarely beat ECFP once evaluation is controlled⁵⁴; we read forward-verification (§5) in that light, narrowly.

Two findings transfer to practice: solve each problem in bounded, frequently-reset contexts with tool access (5% $\rightarrow$ 15%, 1/20 $\rightarrow$ 3/20, directional at $n=20$, McNemar $p \geq 0.25$, §4.3); and use forward-predicted-vs-observed ^{13}C agreement to *re-rank* candidates rather than to decide whether to trust the winner. Match distance failed as an absolute confidence gauge in our selective-prediction test (§5.5, a reported null result), so it does not support abstention thresholds.

8. Limitations

Independence checks. Scoring is mechanical RDKit; solver/verifier runs were transcript-audited *at generation time* for zero web and zero ground-truth access. The committed artefacts are parsed per-compound predictions; transcripts are not deposited (available on request). Closed-book outcomes are separately testable: the formula-only control (§4.6) removes spectra and accuracy collapses; a permutation negative control (§5.5) collapses verification precision from 89.2% to a chance mean of 73.8% (two-sided $p=0.002$ on the full benchmark). Sonnet was comparably recall-bound on a 12-compound subset (recall 33% vs Opus 42% on those compounds), consistent with a task-level recall bound — but within the Claude family, so it speaks to model-instance robustness, not cross-vendor generality.

(i) Pretraining contamination is bounded; we cannot exclude it. Every benchmark compound was mined from open-access literature, so a frontier model may have encountered it in training. The formula-only control (§4.6) bounds pure-recall contribution: masking spectra drops top-1 from 23% to 5%, perfectly nested, McNemar exact $p=0.001$. Of the three formula-only successes, only one (an unlisted silyl aryl sulfonate) has a genuinely determining composition; *N*-tosyl-leucine and [6]-paradol are catalogued compounds for which recall is a live explanation — the bound is unchanged. Accuracy is flat in source publication year across all 194 compounds ($r=-0.007$), the size-adjusted older-versus-newer difference bounded at -5.1 points, 95% CI $[-17.2$ to $+7.0]$. Neither control is randomised; the recency test is not anchored to a *known* training cutoff, since the harness does not disclose one (§9). Replication after a disclosed cutoff remains open.

Neither control addresses verbatim retrieval. The prompt carries spectral strings *exactly as printed in the source article*, because the sampler keeps a compound only when the raw ^1H payload can be recovered verbatim from the open-access full text (§9). Every benchmark compound therefore has its spectrum present verbatim in a document the model may have read — a high-entropy fingerprint. Masking spectra removes chemistry and a retrieval key together: 23% $\rightarrow$ 5% is what reading and retrieval both predict. `scripts/jitter_control.py` builds a paraphrase-invariant set from the released benchmark; we have not run it. It would bound verbatim retrieval only, not recognition from approximate shift patterns.

(ii) Human audit — prepared but not yet run. Elucidation and forward-prediction outputs, and dataset-extraction recall (whether the parser found every IR string), are unvalidated by hand. Transcription fidelity is measured and high (§2.3); record-level recall is not. We have built and frozen a blinded, pre-registered audit package at `data/audit/`, protocol at `docs/EXPERT_AUDIT_PROTOCOL.md`: a difficulty-stratified 30-compound elucidation panel plus seven ranking-only compounds, with a withheld answer key (§4, §5). We have not yet run the panel; until expert results are in, those claims are machine-validated (RDKit InChIKey) but not human-validated. §4 has since reported all 137 analysable misses mechanically (76.6% constitutional isomers, 22.6% scaffold-preserving positional errors); what remains for a chemist is whether a formula-correct, scaffold-wrong candidate is a *chemically reasonable* reading of the spectra.

(iii) Single vendor and an underpowered cross-model comparison. The headline (28.4% top-1, n=194) is Claude Opus on the full benchmark; cross-model evidence (§4.4) is four Claude-family models on a shared 24-compound subset. The ranking is robust (strictly nested, weakest floored at 0%) but no adjacent gap is established at n=24 — quantitative claims hold for the Claude family. §4.7 narrows the cross-vendor question: verification precision exceeds generation recall for Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol on the 60-compound arm, as for Claude; bootstrapping resolves three of four paired gaps and leaves Grok’s gap remains directional at n=60. Those models were reached through a coding-assistant harness with reasoning-effort tiers our arm never had; the tier per arm was not recorded, bounding the recall *ranking* while leaving each arm’s inequality internally valid. The clean-clone contamination control was run for Grok alone.

(iv) Domain-subset scope: the battery-electrolyte case study (§4.5) uses *literature compounds bearing electrolyte-relevant functional chemistry*, not operando or in-cell decomposition spectra, so it demonstrates functional-class transfer of the elucidation bottleneck, not direct assignment of authentic interphase/degradation products, which would require a dedicated operando set.

(v) Constitution-only scoring: correctness is InChIKey connectivity; wrong stereochemistry counts as correct. Full InChIKey scoring gives 21.1% top-1 (§3), 7.3 points lower — an upper bound on full-stereochemistry accuracy. We keep constitution because 1D NMR/IR rarely determines absolute configuration; the 10.3% (20/194) of targets with a defined stereocentre would be penalised for information the prompt never carried.

(vi) Single-sample scoring: each headline compound is scored from one solver prediction set, so reported top-1/recall carry no run-to-run (LLM-sampling) variance estimate; the bootstrap CIs reflect compound sampling only. §5.3 pools ten independent generation passes (recall 32% → 42%), indicating generator stochasticity that single-pass scoring understates.

(vii) Chemical-space coverage: the benchmark draws from open-access organic methodology and total-synthesis literature; it is not representative of organometallics, large biomolecules, or stereochemistry-heavy targets.

(viii) Input regime: IRSpectra-Bench supplies peak lists as transcribed by source-paper authors, not raw instrument files. Concurrent work excludes author-transcribed reports for that reason³⁰ and reports 20.6% top-1 on 34 industrial samples from raw spectra, with stereochemistry retained. Our comparable figure is full-InChIKey 21.1%, not the 28.4% constitution-only headline (§3). The two settings measure different things: reading the published record across thousands of laboratories, and inference from instrument output in one. Neither bounds the other.

9. Methods

Mining and resolution. PMC-OA full text was fetched from `s3://pmc-oa-opendata`, parsed deterministically, and resolved with OPSIN, RDKit and SELFIES, with an optional cached PubChem fallback.

Benchmark and agents. Problems were sampled from `irexp_resolved`, stratified by RDKit ring analysis, and de-duplicated across rounds by InChIKey. A compound was admitted only if its raw ¹H payload — multiplicities and *J* values as printed — could be re-extracted verbatim from the PMC-OA full text of its source article (`benchmark_v2.raw_1h_for`), which puts genuine coupling information in the prompt and, unavoidably, the source article’s exact string (§8). The battery-electrolyte

subset (§4.5) was drawn by SMARTS filters for six electrolyte functional classes (carbonate, sulfonfyl/sulfonate, nitrile, sp³-C-F, phosphoryl, glyme/oligoether), balanced to eight compounds per class (48 curated; 46 scored after two yielded no parseable candidate), excluding compounds used elsewhere; it was *J*-enriched and spectrally validated. Solver and forward-prediction agents were independent Claude-Opus sub-agents under the same subscription, instructed closed-book and audited for zero web/answer access. Scoring used RDKit InChIKey-connectivity match; similarity Morgan(2, 2048) Tanimoto; forward-verification a symmetric chamfer distance over ¹³C peak sets. The core protocol trains no model and uses no paid API; the §5.6 generator and the §5.4 learned ¹³C verifier are the only trained components, reported as complements and fenced from the headline results.

Models and versions. All experiments used Anthropic Claude models via the consumer subscription (claude.ai). The §4.4 comparison spanned, in capability order, Claude Haiku, Claude Sonnet, Claude Opus and Claude Fable 5; the headline benchmark (§4.1) and forward-verification (§5) used Claude Opus. `docs/MODELS.md` records, per experiment, the model, data directory, collection date, harness and tool access, and scoring code path, across three dated windows:

- **2026-06-09 to 2026-06-11** (UTC) — every candidate structure behind the headline results: §4.1, §4.3–§4.5, and the candidate pools all of §5 re-ranks.
- **2026-07-28** — the formula-only contamination control (§4.6), re-solving the same 60 compounds with spectra masked and generating its own candidates; it touches Table 5 alone.
- **2026-08-07** — three collections forward-predicting ¹³C for candidates the June run had already produced (the §5.2 extension to all 194 compounds, the §5.3 coverage-gap closure, and the §5.6 re-run); none introduced a new candidate structure, and no recall number in the paper changes.

One *verified* number does: the §5.6 re-run supersedes a top-1 whose original forward-prediction outputs were never deposited and are lost, disagreeing with the figure it replaces (47% against 41%); §5.6 reports it as a re-run rather than a reproduction.

No model snapshot can be reported. The consumer harness exposes no checkpoint identifier and records nothing about which build served a given request. It exposes no decoding parameters either, and none were set or recorded. `Claude Opus 4.8` is evidenced only for the 2026-06-09 pilot, so a mid-window build change cannot be excluded, and we do not claim the same build served the main round two days later. Re-running reproduces the protocol distributionally rather than exactly, and exact agreement is not offered as a target. Fixed are the dated collection windows, the frozen per-compound outputs, and mechanical scorers that regenerate every number from them: exact reproducibility of scoring and analysis, not of inference (`docs/MODELS.md` §7).

Reproducibility. Every round is frozen: questions, ground-truth answers, per-agent raw outputs, predictions and scorer outputs are released; the sampler, scorer and forward-verification harness are scripted end-to-end.

Supporting Information figures

These are supplied as a separate Electronic Supplementary Information document (`docs/paper_esi.pdf`, built by `scripts/build_pdf.py`), as RSC requires; they are listed here for reference.

- **Fig. S1** (docs/figures/fig0_overview.png) — study design: open multimodal data (IRexp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free core pipeline.
- **Fig. S2** (docs/figures/fig4_dataset.png) — IRexp composition: IR records → NMR-paired → structure-linked → full IR + ^1H + ^{13}C + structure quadruples.
- **Fig. S3** (docs/figures/fig2_size.png) — accuracy vs molecular size (heavy-atom bucket); the monotonic 60% → 7% top-1 gradient.
- **Fig. S4** (docs/figures/fig6_electrolyte.png) — top-1 and recovered accuracy on IRSpectra-Bench-Electrolyte by battery-electrolyte chemical class (n=46): $\text{sp}^3\text{-C-F}$ easiest (50%), sulfonyl and nitrile hardest (12%); overall 26%/28%.
- **Fig. S5** (docs/figures/fig_generator_probe.png) — trained-generator probe (§5.6; complement, not training-free protocol): generation recall and deterministic-HOSE top-1 on the 194-compound benchmark for Claude / + scaffold enumeration / + trained generator. Enumeration’s near-degenerate isomers collapse the verifier (28.4 → 16.0%) while the generator’s formula-correct candidates convert (28.4 → 35.1%).
- **Fig. S6** (docs/figures/fig_verifier.png) — learned-verifier probe (§5.4; complement, not training-free protocol). (A) Conditional-on-recall top-1 (n=65, whole benchmark) for four verifiers: a GNN trained on the same nmshiftb2 data as the HOSE lookup reaches the LLM verifier’s level (91% vs 89%) where the lookup (85%) does not move off the solver’s own ranking. (B) Held-out ^{13}C MAE — the learned model is $\approx 2\times$ sharper (1.70 vs 3.23 ppm).

Author contributions

I.Y.: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing — original draft. **R.S.:** methodology, software, formal analysis, investigation, validation (trained-generator and learned-verifier probes, §5.4 and §5.6), writing — review and editing. **R.A.V.-H.:** conceptualization, methodology, supervision, writing — review and editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

Table 11 lists every released component and the script that regenerates it, all of them in the project repository. The archival deposit — a complete frozen snapshot of dataset, benchmark, answer keys, predictions, scripts, figure regeneration and the expert-audit package — is deposited on Zenodo, and its DOI is supplied at proof stage; GitHub is the development mirror and the Zenodo record the citable version.

891 **Table 11. Released artefacts, and the data and code behind each one.**

component	data and code
IRexp and its structure-complete split benchmark rounds and the within-compound control ground-truth integrity audit	data/irexp/, data/irexp_resolved/ — spectro_scraper/ (mining pipeline) data/benchmark*/ — scripts/benchmark_v2.py
headline accuracy (Table 2) battery-electrolyte subset (§4.5)	data/benchmark*/clean_qids.json — scripts/validate_benchmark.py scripts/score_main.py data/benchmark_electrolyte/ — scripts/build_electrolyte_bench.py, scripts/score_electrolyte.py
formula-only contamination control (§4.6)	data/modality/ — scripts/modality_ablation.py
publication-recency control (§4.6)	data/audit/recency_control.json — scripts/contamination_recency.py
forward-verification, original arm (§5.2) ...extended to all 194 compounds generate-wide arm (§5.3)	data/fverify/ — scripts/forward_verify.py data/fverify_main/ — scripts/forward_verify_main.py, scripts/forward_verify_all.py data/gw/, data/fverify2/ — scripts/score_generate_wide.py, scripts/ladder_significance.py
...its coverage gap, closed non-LLM verifier comparison (Table 9)	data/fverify_gw/ — scripts/forward_verify_gw.py data/fverify/hose_results.txt, data/fverify/verifier_table_results.txt — scripts/hose_predict.py (incl. coverage), scripts/verifier_table.py, scripts/verifier_leakage.py
negative control and selective prediction (§5.5)	scripts/verifier_diagnostics.py
what a miss is, and isomer separability (§4.1, §5.1)	scripts/analyze_misses.py, scripts/isomer_separability.py
difficulty-threshold sensitivity (§3)	scripts/difficulty_sensitivity.py
licence-pool split (§2.1)	scripts/split_license_pools.py
recall headroom and scaffold enumeration	scripts/analyze_recall_headroom.py, scripts/enumerate_isomers.py, scripts/closing_the_gap.py
blinded expert-audit package (§8)	data/audit/ — scripts/make_audit_sample.py
corpus reweighting (§3)	scripts/corpus_reweight.py
cross-vendor recall at matched budget, and the paired gap (§4.7)	data/cross_vendor/ — scripts/cross_vendor_budget.py, scripts/cross_vendor_gap.py
ring-system names in peak assignments (§3)	scripts/prompt_leakage.py
paraphrase-invariant benchmark, for the control not yet run (§8)	— (regenerated, not deposited) — scripts/jitter_control.py

component	data and code
manuscript integrity gates	<code>scripts/check_manuscript.py</code> , <code>scripts/verify_statistics.py</code> , <code>scripts/check_compression.py</code> , <code>scripts/check_layout.py</code>

Trained complements, both fenced from the training-free protocol. The §5.6 generator ships as a self-contained bundle at `contrib/generator_probe/` — candidates, the de-leaked split with its InChIKey-14 manifest, the verification scripts (`scripts/closing_the_gap_gen.py`, `scripts/forward_verify_gen.py`, `scripts/verify_leakage_exact40.py`) and the blind forward-prediction outputs behind its verified top-1 (`data/fverify_gen/`), so its scorer runs with no missing predictions. The §5.4 learned verifier ships `scripts/gnn_predict.py` (extract / train / score / control), the trained model `data/nmrshiftdb/gnn_c13.pt`, per-compound results and both leakage checks (`data/fverify/gnn_results.txt`), and the write-up `docs/VERIFIER_PROBE.md`. Model checkpoints are deposited on Zenodo. Both trained predictors of §5.4 need the `nmrshiftdb2` dump, which we cannot redistribute; the README gives the one-line fetch.

Companion technical notes: `docs/BENCHMARK.md`, `docs/FORWARD_VERIFY.md`, `docs/MODALITY_ABLATION.md`, `docs/EXPERT_AUDIT_PROTOCOL.md`, `docs/MODELS.md`.

Two cautions for re-users. The public `irexp_release/train` split does not hold out IR Spectra-Bench — it overlaps by 117/200 InChIKey-14 — so de-leak with `contrib/generator_probe/build_exp_manifest.py` before training anything that will be evaluated on the benchmark. And the held-out answer keys `data/audit/key.jsonl` and `data/modality/key.json` are deposited but flagged *withhold from blinded reviewers*.

Licensing and attribution

IRexp is derived from two open-access source pools and redistributed under terms compatible with each (see `data/NOTICE`). **(a) Redistribution:** we release only *extracted numeric data* — IR band lists, $^1\text{H}/^{13}\text{C}$ shift lists, and resolved structures (SMILES/SELFIES/InChIKey) — plus each record’s source DOI/accession, not source full text, figures, or PDFs. **(b) Two separable pools:** 119,345 records derive from the PMC Open-Access Subset (CC-BY-4.0) and 1,888 from the Chemotion RADAR4Chem FT-IR deposit (CC-BY-SA-4.0). The two are separable losslessly by `source_doi` — Chemotion records carry the 10.22000 prefix — and `scripts/split_license_pools.py` materialises them as two files with each record stamped license, so users may take the CC-BY pool alone; any combined or Chemotion-derived release carries CC-BY-SA-4.0 to honour the ShareAlike term. Code is released under the MIT License. **(c) Attribution:** re-users must cite this dataset (Zenodo DOI above) and attribute the original publications via each record’s `source_doi`.

Acknowledgements

Use of AI tools

This work studies a large language model, and LLMs were also used as instruments and as writing aids; we state both roles explicitly. **As an object of study:** all reported elucidation, forward-prediction and verification results were produced by Claude models invoked under the protocol of §3 and §9 — these are the measurements the paper reports, not assistance in producing it. **As**

a writing aid: the authors used an LLM-based coding assistant for figure generation, analysis scripting, manuscript copy-editing, and for the internal review pass that produced several of the corrections recorded in the repository history. No text, number, figure or citation in this manuscript was accepted without author verification against the released data and code; every quantitative claim regenerates from the scripts in `scripts/`. The authors take full responsibility for the content.

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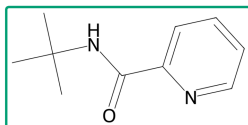
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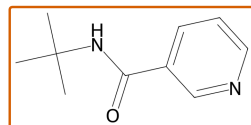
Table of contents entry

28% top-1 on blind, real IR + NMR spectra

observed ^{13}C , $\text{C}_{10}\text{H}_{14}\text{N}_2\text{O}$



0.42 ppm selected



1.30 ppm rejected

recall (34%), not verification (89%), is the wall

Recall, not verification, is the wall. On real, blind IR + ^1H + ^{13}C literature spectra a frontier LLM proposes the true structure for 34% of 194 compounds and, once proposed, selects it 89% of the time — lifting top-1 from 28% to 30%.